IE 376

PRODUCTION INFORMATION SYSTEMS

Course Notes

Lecture # 9

Theory of Constraints

The Goal - Synchronous Operations

- Optimized Production Technology (OPT) developed by Eliyahu Goldratt
  - **The Goal**: Goldratt illustrated the concepts of OPT in the form of a novel.
  - **The Race**: written to overcome difficulties encountered in the implementations.
- **THE GOAL OF A FIRM IS TO MAKE MONEY**
  (As per Goldratt - making money allows guaranteed long-term survival and prosperity. With money, other goals can be achieved.)

- To ensure that all elements are working in harmony (i.e. synchronously) to achieve the goal of a firm.
- Emphasis is on total system performance - not on such local measures as labor or machine utilization or efficiency. The sum of local optima is not equal to the optimum of the whole.

Operational Measurements

1. Throughput - the rate at which money is generated by the system through sales
2. Inventory - all the money the system has invested in purchasing things it intends to sell
3. Operating expenses - all the money that the system spends to turn inventory into throughput

How to Achieve the Goal?

Increase throughput while simultaneously reducing inventory and reducing operating expenses.

What can limit you from achieving the goal?

Constraints!! Bottlenecks!!

Theory of Constraints (Goldratt)

- A system's constraint is anything that limits the system from achieving a higher performance with respect to its its goal
- Any system has very few constraints
- Any system must have at least one constraint
  - The existence of constraints represents opportunities for improvement.

Types of Constraints

- Market Constraints
- Material Constraints
- Capacity Constraints
- Logistical Constraints
- Managerial Constraints
- Behavioral Constraints
  - Tell me how you measure me, and I will tell you how I will behave. If you measure me in an illogical way... do not complain about illogical behavior.

Optimized Production Technology (OPT)

OPT is an analytical technique that is designed to achieve the **goal** through realistic *optimized* schedules.

Known to involve four basic stages

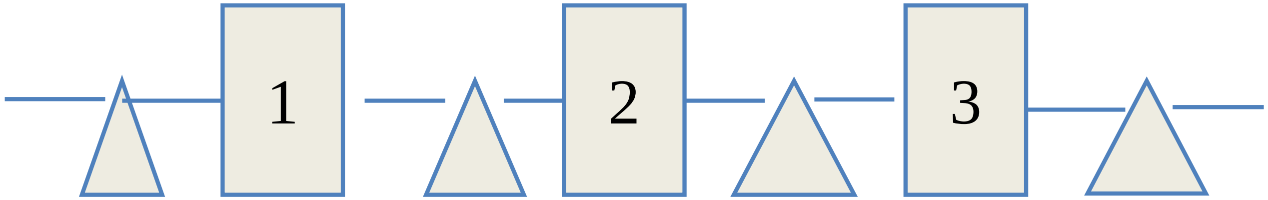
- Determine the bottleneck resource in the shop
- Schedule to use the bottleneck resource most effectively.
- Schedule the remainder of the shop up to the bottleneck.
- Schedule the remainder of the shop after the bottleneck.

Bottlenecks

- A bottleneck could be a machine whose capacity limits the throughput of the whole production process.

- Finding the bottleneck
  - Run a capacity resource profile (compute work load as a percentage of capacity for each resource).
  - The standard data used may not be accurate and hence visit the facility and talk to workers to confirm or modify the profile.
  - Identify the bottleneck.

3-Machine Flow shop



Throughput
30 parts/hr

Production rates

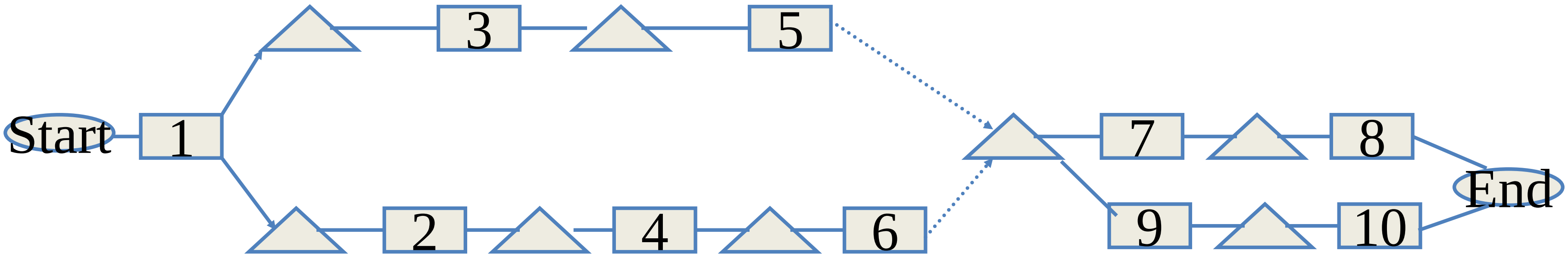
- Work station 1: **60** parts/hr
- Work station 2: **30** parts/hr
- Work station 3: **40** parts/hr

OPT Philosophy

- Rule 1. The level of utilization of a non-bottleneck is determined not by its own potential, but by some other constraint in the system.

- Rule 2. Utilization and activation of a resource are not synonymous.
  - Efficiency versus Effectiveness

Applying OPT:



- Bottleneck: 3, Nonbottleneck: 5
- Bottleneck: 5, Nonbottleneck: 3
- Bottleneck: 5, Nonbottleneck: 6
- Bottleneck: 8, Nonbottleneck:10

Examples from “The Goal” (great novel – read it !)

- “Keep the robots working so that they pay for themselves”

- *Logic:* robots are very expensive… In order to recover your investment, you should keep them busy all the time…

- *The problem with the statement:* unless you actually sell the finished products, you are only building inventory! Keeping the equipment and people busy does not mean that you are making more money.

A) Agree B) Disagree

Examples from "The Goal"

"A plant in which everyone is working all the time is very inefficient"

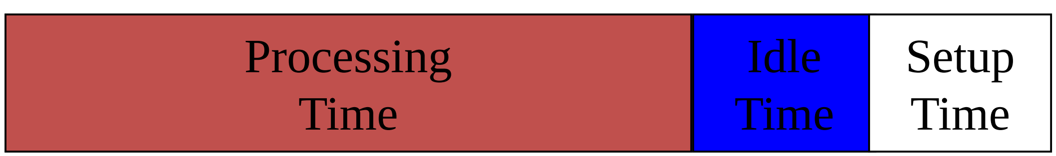
The logic of the statement: Work content may fluctuate over time… leaving some people idle from time to time. Furthermore, some production stages may be very slow (bottleneck), and others very fast (non-bottleneck). Non-bottlenecks cannot by definition work as hard as the bottleneck (without doing harm).

A) Agree B) Disagree

Setup Times

- Bottlenecks

- Non-bottleneceks



OPT Rules

- Rule 3. An hour lost in the bottleneck is an hour lost for the total system.
- Rule 4. An hour saved at a non-bottleneck is just a mirage.
  - Variable batch sizes
- Rule 5. Bottlenecks govern both throughput and inventory in the system.

Managing Bottleneck Capacity– Theory of Constraints

Determine bottleneck work centers

- Rough-cut capacity planning
- Capacity resource planning

Look for quick solutions to eliminate bottlenecks

- Expand capacity
- Alternate routings

Concentrate scheduling efforts on managing bottleneck resources

- Schedule jobs that run through bottlenecks separately from non-bottleneck jobs
- Use finite scheduling for bottleneck jobs, with horizontal loading and back scheduling for the most critical

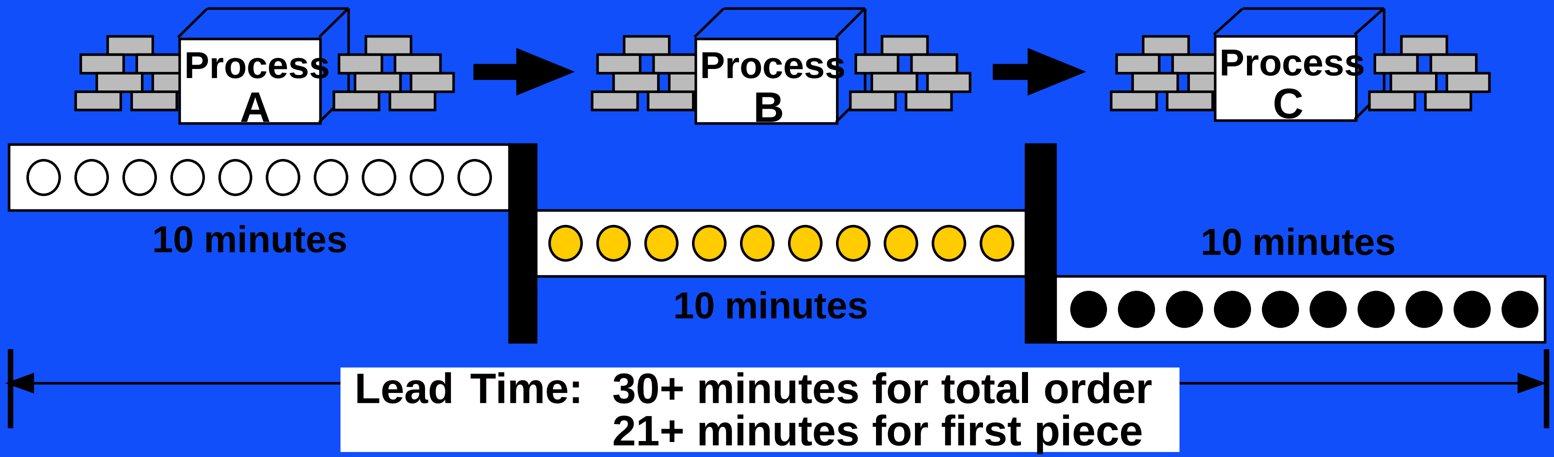
Lot Sizes

- The transfer batch: the lot size from the parts point of view
- The process batch: the lot size from the resource point of view

- Rule 6. Transfer batch may not, and many times should not, be equal to the process batch.
  - **Lot splitting**

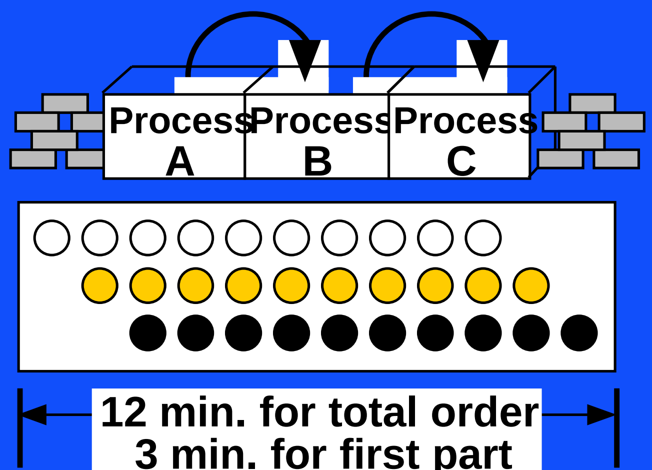
IMPACT OF BATCH SIZE REDUCTION

• Batch & Queue Processing



• Continuous Flow Processing

Treatment of Lot Sizes



Process Batch Versus Transfer Batch

- Dedicated Assembly Line: *What should the batch size be?*

- Process Batch
  - Related to length of *setup*.
  - The longer the setup the larger the lot size required for the same capacity.

- Transfer Batch *Why should it equal process batch?*
  - The smaller the transfer batch, the shorter the cycle time.
  - The smaller the transfer batch, the more material handling.
  - Tradeoff?

TOC and Lot Sizing

- Rule 7. The process batch should be variable, not fixed.
- Lot sizes are calculated differently for bottleneck and non-bottleneck resources
  - For the same finished item, lot sizes at different operations may be different
  - TOC splits orders at non-bottleneck resources and combines orders at bottlenecks
- This maintains supply of non-bottleneck parts while reducing setup time and/or increasing efficiency at the bottleneck

Lead-Time Management

- Actual lead times are not fixed.
- Lead times are not known a priori, but depend upon the sequencing at the limited capacity or bottleneck resources.

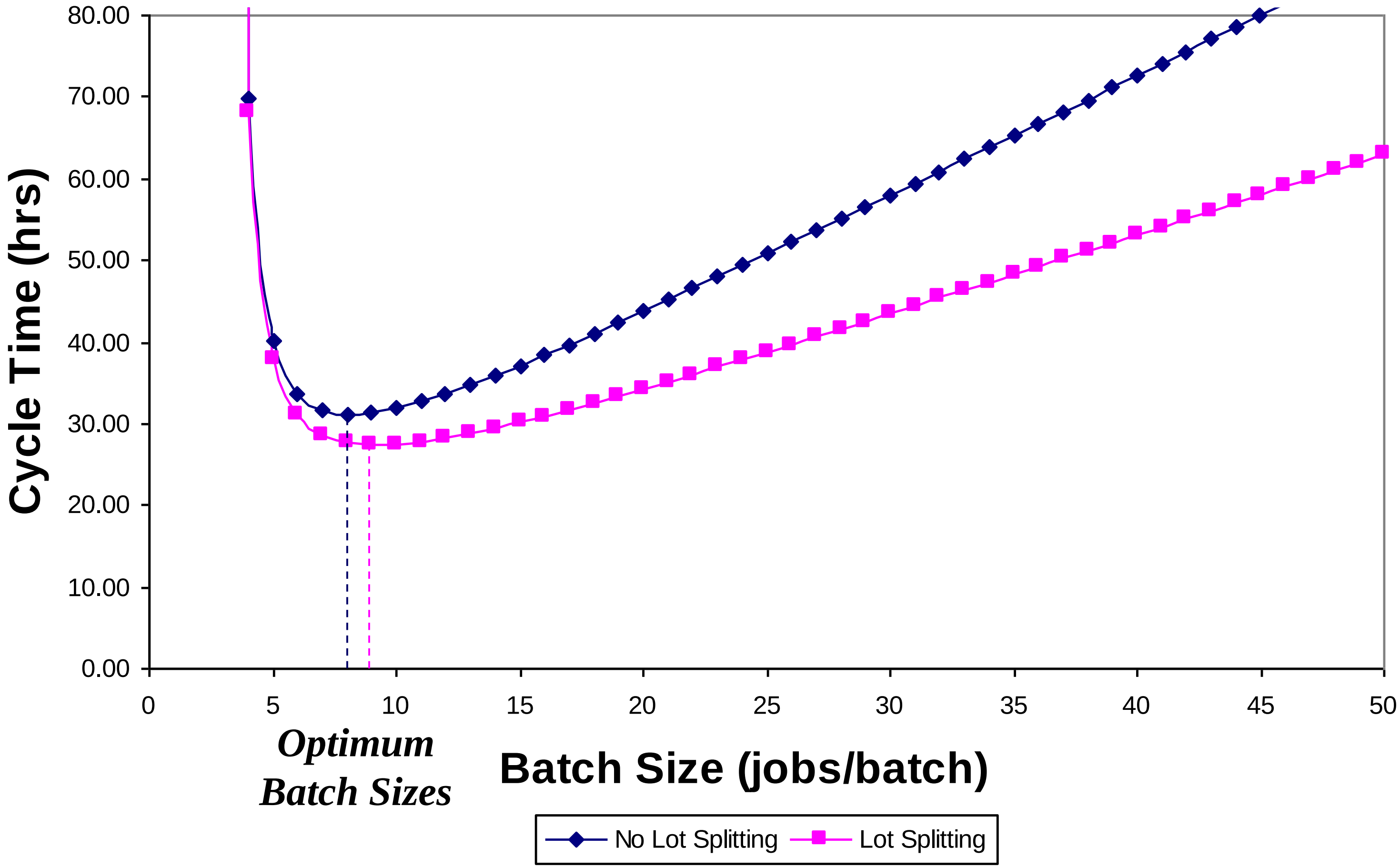
*The average **Cycle time** at a station =*

Move time + Queue time +

Setup time + Process time +

[wait-to-batch time + wait-in-batch time + wait-to-match time]

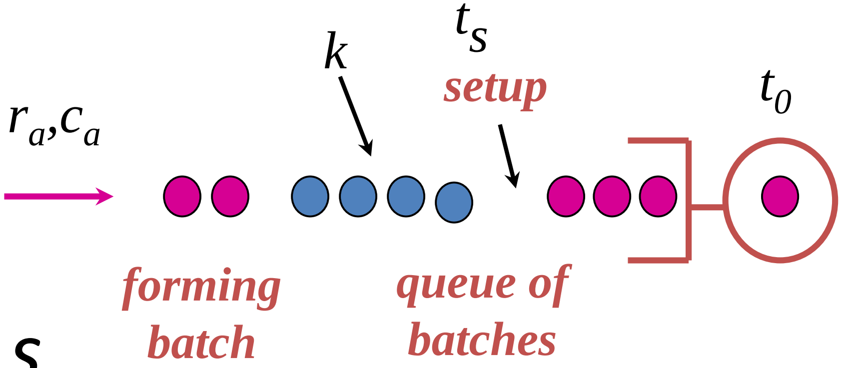
Cycle Time vs. Batch Size



Serial Batching

- Parameters:

$$k = \text{serial batch size } (10)$$
$$t = \text{time to process a single part } (1)$$
$$s = \text{time to perform a setup } (5)$$
$$c_e = \text{CV for batch (parts + setup) } (0.5)$$
$$r_a = \text{arrival rate for parts } (0.4)$$
$$c_a = \text{CV of batch arrivals } (1.0)$$

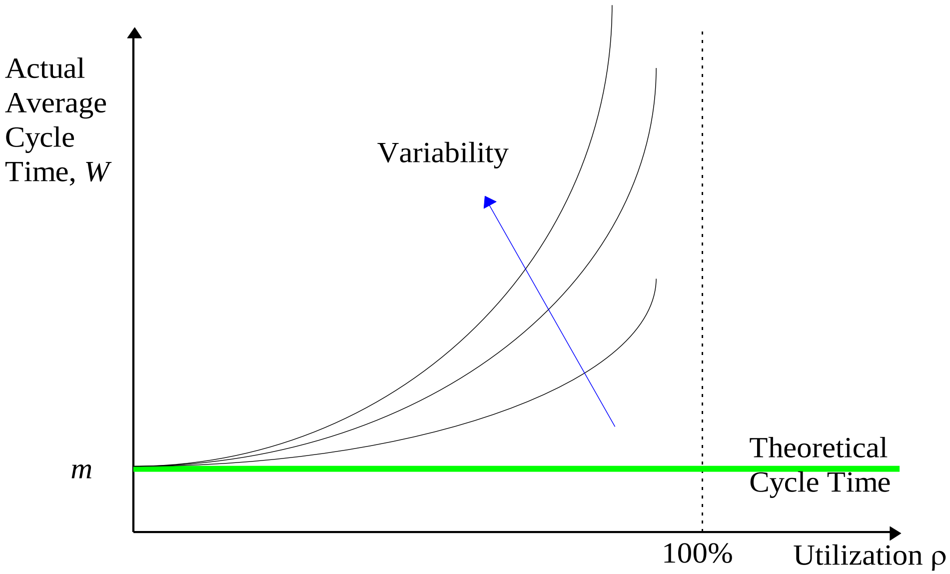


- Time for batch: $t_e = kt + s$

  (t_e = 15)

Queuing Theory: Variability + Utilization = Waiting

- Throughput-Delay curve:



$$\mathrm{CT}_q \approx V \times U \times t$$

$$\approx \left(\frac{c_a^2 + c_e^2}{2}\right)\left(\frac{u}{1-u}\right) t_e$$

variability effect

utilization effect

mean service time

Observations

CT_q = the expected waiting time spent in queue.

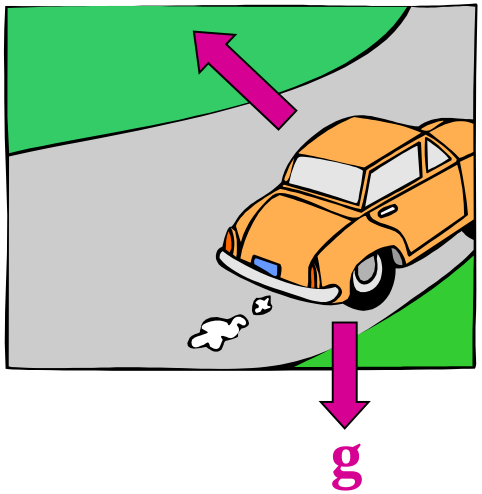
– Separate terms for variability, utilization, process time.

– Flow variability, process variability, or both can combine to inflate queue time.

– ***Variability causes congestion!***

delay times typically make up 90% of CT

Probabilistic Intuition



Uses of Intuition:

- driving a car
- throwing a ball

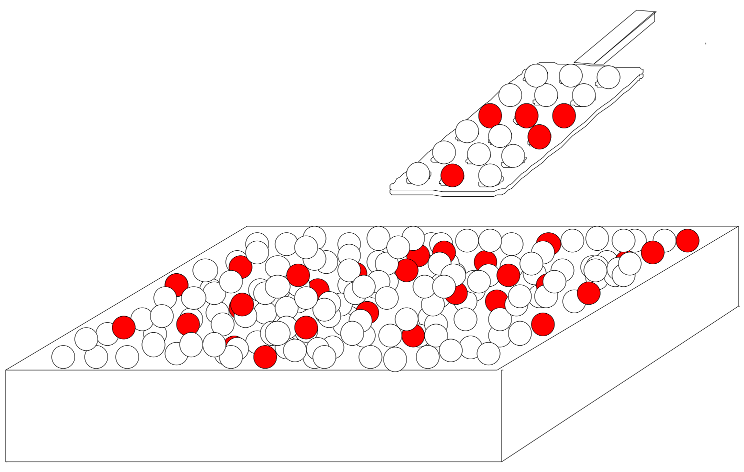
First Moment Effects:

- Throughput increases with machine speed
- Throughput increases with availability
- Inventory increases with lot size
- Our intuition is good for first moments

Probabilistic Intuition (cont.)

Second Moment Effects:

- Which is more variable – processing times of parts or batches?
- Which are more disruptive – long, infrequent failures or short frequent ones?
- Our intuition is less secure for second moments
- Misinterpretation – e.g., regression to the mean

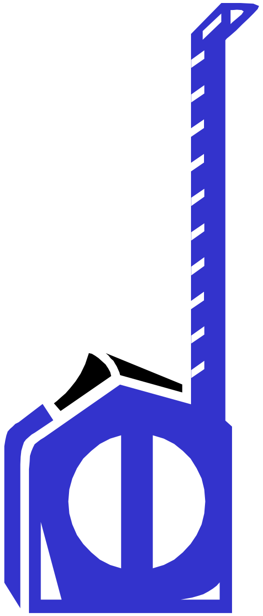


Measuring Process Variability

$$t_e = \text{mean process time of a job}$$

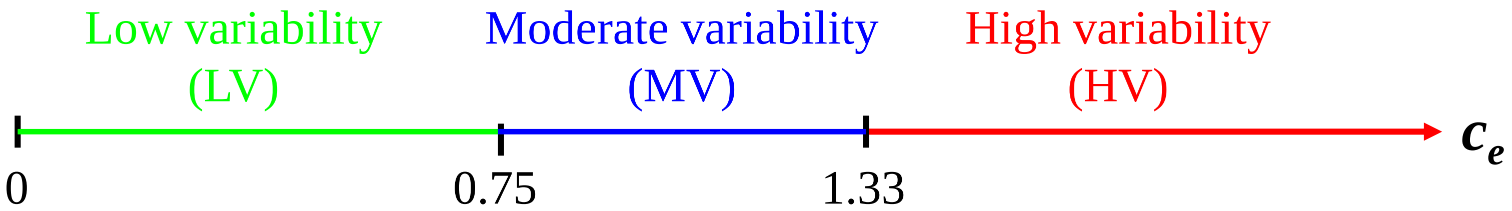
$$\sigma_e = \text{standard deviation of process time}$$

$$c_e = \frac{\sigma_e}{t_e} = \text{coefficient of variation, CV}$$



Note: we often use the "squared coefficient of variation" (SCV), c_e^2

Variability Classes



Effective Process Times:

- ***actual*** process times are generally LV
- ***effective*** process times include setups, failure outages, etc.
- HV, LV, and MV are all possible in effective process times

Measuring Process Variability – Example

Trial	Machine 1	Machine 2	Machine 3
1	22	5	5
2	25	6	6
3	23	5	5
4	26	**35**	**35**
5	24	7	7
6	28	**45**	**45**
7	21	6	6
8	30	6	6
9	24	5	5
10	28	4	4
11	27	7	7
12	25	**50**	**500**
13	24	6	6
14	23	6	6
15	22	5	5
t_e	25.1	13.2	43.2
s_e	2.5	15.9	127.0
c_e	0.1	1.2	2.9
Class	LV	MV	HV

Managing the Flow Rate

- Increase Throughput
  - Manage supply and demand
- Increase Process Capacity
  - Reduce starvation (synchronize flows better)
  - Reduce blockage (appropriate select buffer sizes)
- Increase Effective Capacity
  - Reduce resource unavailability (maintenance policies, better HR management, less frequent setup/changeover, reduced setup/changeover times)
- Increase Theoretical Capacity

Increasing Theoretical Capacity

$\sum$ (1/Unit Load) x Load Batch x Sched. Availability

- Decrease the unit load on *bottleneck* resource pool
  - Work smarter (remove non-value-adding activities, e.g., reduce/eliminate setups and changeovers)
  - Work faster (acquiring faster equipment, increase resource allocation, offering incentive to increase output)
  - Do it right the first time
  - Change product mix (processing more products that require less time)
  - Move work content from bottlenecks to non-bottlenecks or third-party
- Increase the load batch of *bottleneck* resources
- Increase the number of resource units in the *bottleneck* resource pool (investment decisions)
- Increase scheduled availability of *bottleneck* resource pool (work longer)

Bottlenecks, Non-Bottlenecks and Capacity Constrained Resources

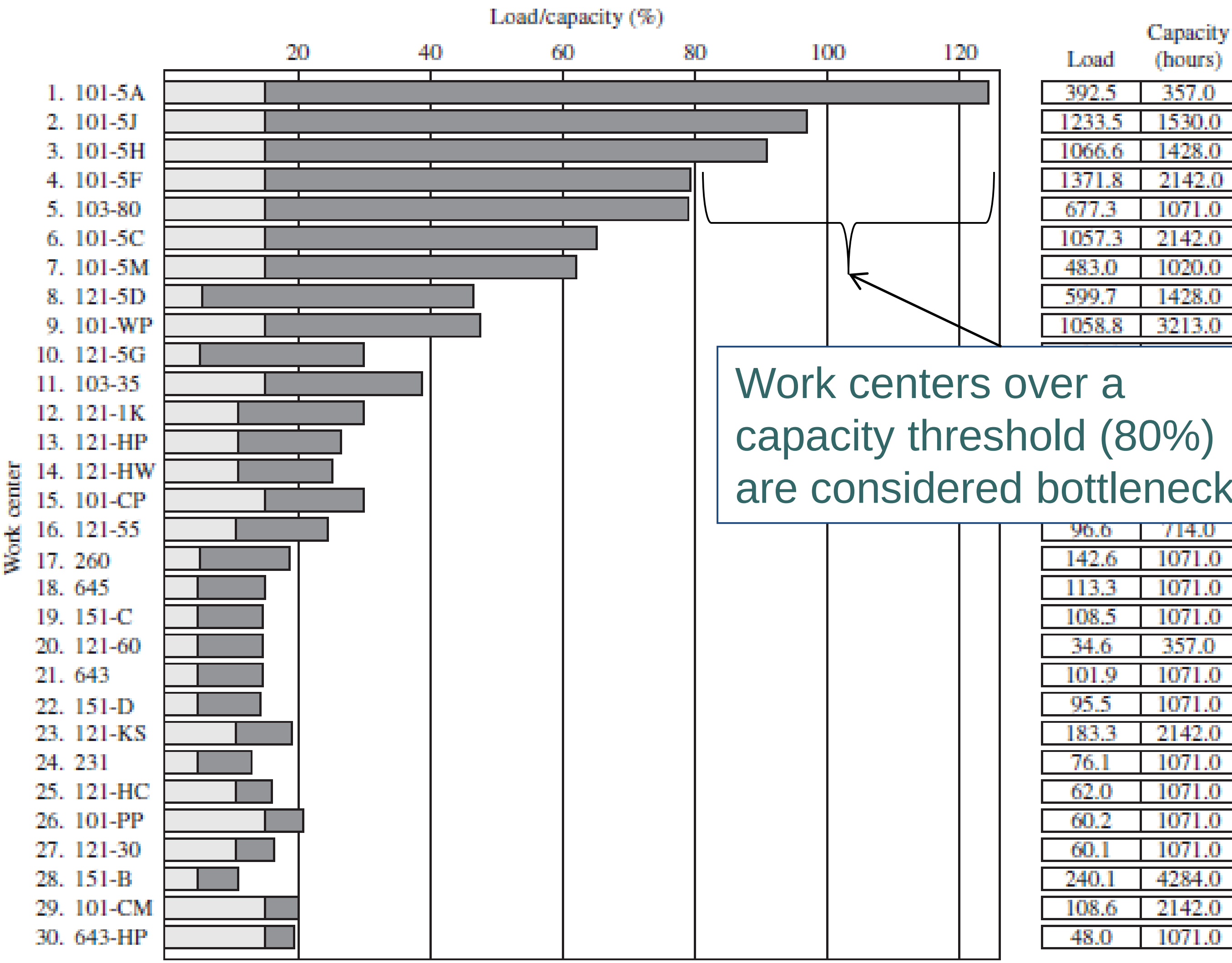
- Capacity is defined as the available time for production excluding maintenance and other downtime.

- A bottleneck is a resource whose capacity is less than the demand placed on it. Bottleneck limits the throughput.

- A non-bottleneck is a resource whose capacity is greater than the demand placed on it.

- A capacity constrained resource is one whose utilization is close to capacity and could be a bottleneck if it is not scheduled carefully.

Capacity Utilization Chart



Work centers over a capacity threshold (80%) are considered bottlenecks

Theory of Constraints Scheduling

- Drum-Buffer-Rope (DBR) Scheduling
  - Drum–bottleneck work centers which control the tempo of workflow through the plant
  - Buffer–inventory and/or scheduling activities to protect the throughput at bottlenecks from random variation
  - Rope–use of pull scheduling at non-bottleneck resources

- Material moves through non-bottleneck resources as quickly as possible, bottlenecks are managed for maximum efficiency
  - After bottleneck, jobs are pushed to the end of line as quickly as possible

Buffers

- TOC uses both safety stock and safety lead time at bottleneck operations
  - **Safety lead time** is introduced between sequential orders at the bottleneck.
    - Jobs are released at the bottleneck rate but far enough in advance so that a time buffer is maintained before bottleneck. (Jobs pulled from first stage by bottleneck)
  - **Safety stock** of completed parts from preceding, non-bottleneck operations is held in front of the bottleneck to prevent shortages.
    - Buffer set before bottleneck so that it keeps up with the beat and never starves (even if a nonbottleneck is down)

Cycle Time

Definition (Station Cycle Time): ***The average cycle time at a station is made up of the following components:***

> ***cycle time = move time + queue time + setup time + process time + wait-to-batch time + wait-in-batch time + wait-to-match time***

delay times typically make up 90% of CT

Definition (Line Cycle Time): ***The average cycle time in a line is equal to the sum of the cycle times at the individual stations less any time that overlaps two or more stations.***

Reducing Queue Delay

$$CT_q = V \times U \times t$$

$$\left(\frac{c_a^2 + c_e^2}{2}\right) \qquad \left(\frac{u}{1-u}\right)$$

Reduce Variability

- failures
- setups
- uneven arrivals, etc.

Reduce Utilization

- arrival rate (yield, rework, etc.)
- process rate (speed, time, availability, etc)

Variability

Definition: Variability is anything that causes the system to depart from regular, predictable behavior.

Sources of Variability:

- setups
- machine failures
- materials shortages
- yield loss
- rework
- operator unavailability
- workpace variation
- differential skill levels
- engineering change orders
- customer orders
- product differentiation
- material handling

Reducing Batching Delay

$$CT_{batch} = \textit{delay at stations} + \textit{delay between stations}$$

Reduce Process Batching

- Optimize batch sizes
- Reduce setups
  - Stations where capacity is expensive
  - Capacity vs. WIP/FT tradeoff

Reduce Move Batching

- Move more frequently
- Layout to support material handling (e.g., cells)

Reducing Matching Delay

CT_{batch} = *delay due to lack of synchronization*

Reduce Variability

- High utilization fabrication lines
- Usual variability reduction methods

Improve Coordination

- scheduling
- pull mechanisms
- modular designs

Reduce Number of Components

- product redesign
- kitting

Increasing Throughput

$$TH = P(\text{bottleneck is busy}) \times \text{bottleneck rate}$$

Reduce Blocking/Starving
- buffer with inventory (near bottleneck)
- reduce system "desire to queue"

$$CT_q = V \times U \times t$$

Reduce Variability

Reduce Utilization

Increase Capacity
- add equipment
- increase operating time (e.g. spell breaks)
- increase reliability
- reduce yield loss/rework

Note*: *if WIP is limited, then system degrades via TH loss rather than WIP/CT inflation

Improving Customer Service

$$LT = CT + z\,\sigma_{CT}$$

Reduce CT Visible to Customer

- delayed differentiation
- assemble to order
- stock components

Reduce Average CT

- queue time
- batch time
- match time

Reduce CT Variability

generally same as methods for reducing average CT:

- improve reliability
- improve maintainability
- reduce labor variability
- improve quality
- improve scheduling
- etc…

Corrupting Influence Takeaways

Variance Causes Congestion:

- many sources of variability
- planned and unplanned

Variability and Utilization Interact:

- congestion effects multiply
- utilization effects are highly nonlinear
- importance of bottleneck management

Corrupting Influence Takeaways (cont.)

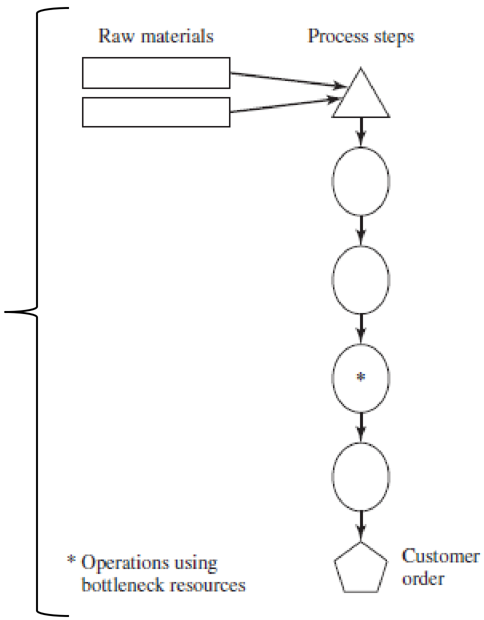
Variability Propagates:

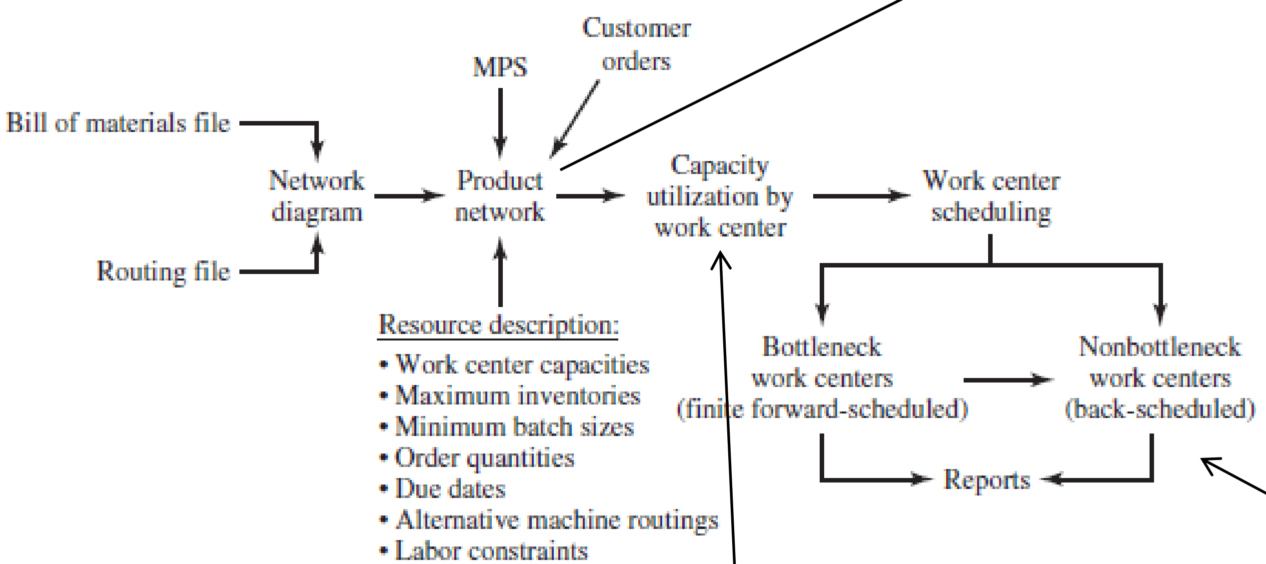
- flow variability is as disruptive as process variability
- non-bottlenecks can be major problems

Variability *Must* be Buffered:

- WIP
- capacity
- demand

TOC Scheduling





Potential bottlenecks are identified using rough-cut capacity planning

Scheduling is completed according to the work center type

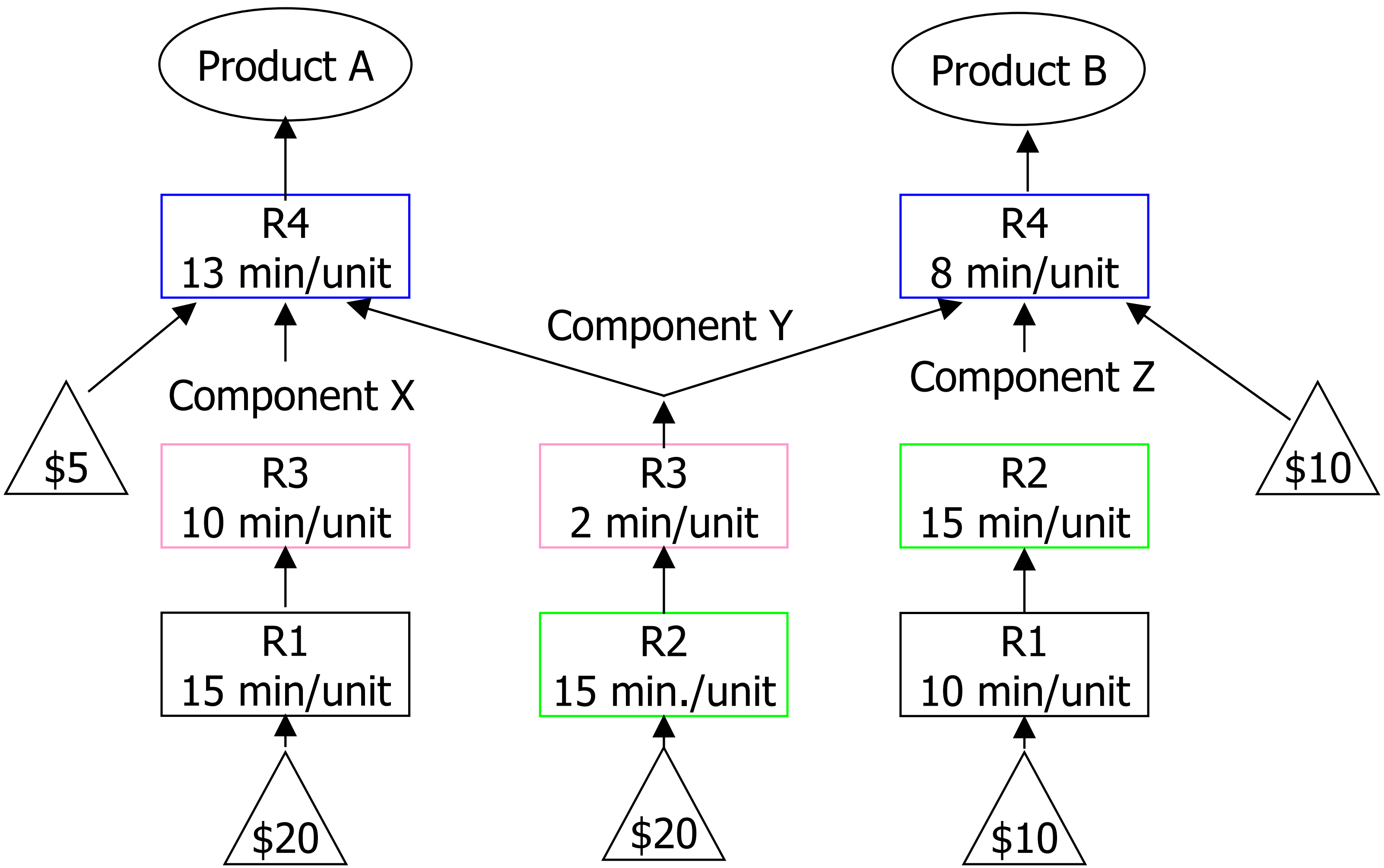
Example: What to Produce?

- Which product should you produce?

	Product A	Product B
Selling price	$90	$100
Material cost	$45	$40
Labor time per unit	55 min.	50 min.
Market demand	unlimited	unlimited

- Total available hours is 40 hours per week
- Weekly operating expenses are $5000
- Product B??? *B has a higher selling price, a lower material cost, and requires less labor time!*

Routing and Processing Information



Analyzing Production Capability

Resource	**Product A** Processing Time (min)	**Product A** Max. Output (2400 min/week)	**Product B** Processing Time (min)	**Product B** Max. Output (2400 min/week)
R1	15	160	10	240
R2	15	160	30	80
R3	12	200	2	1200
R4	13	184	8	300

Produce 80 units of B	80x$60=$4800 per week	$200 loss per week
Produce 160 units of A	45x160=$7200 per week	$2200 profit per week

Modeling Matters!

I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind; it may be the beginning of knowledge, but have scarcely, in your thoughts, advanced to the stage of Science, whatever the matter may be.

- Lord Kelvin

Why Models?

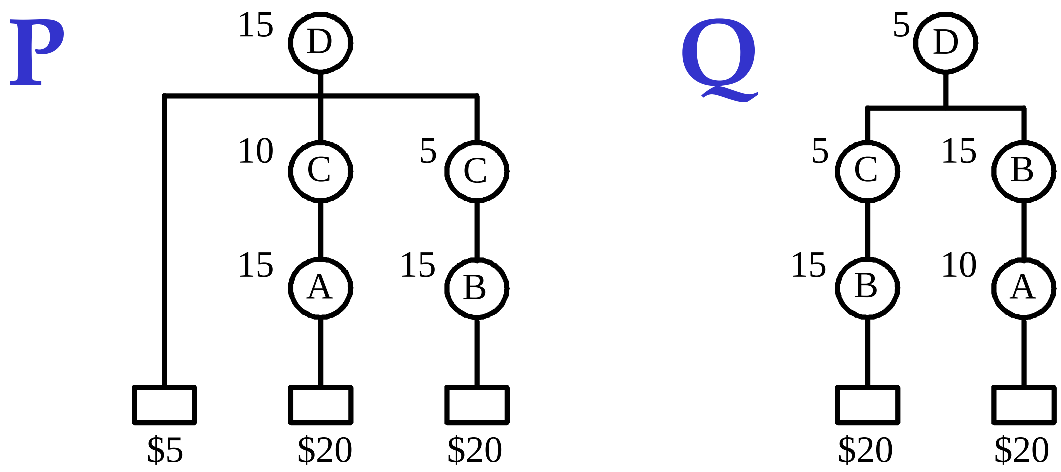
State of world:

- Data (not information!) overload
- Reliance on computers
- Allocation of responsibility (must justify decisions)

Decisions and numbers:

- Decisions are numbers
  - How many distribution centers do we need?
  - Capacity of new plant?
  - No. workers assigned to line?
- Decisions depend on numbers
  - Whether to introduce new product?
  - Make or buy?

Goldratt Product Mix Problem



Product	P	Q
Price	$90	$100
Max Weekly Sales	100	50

- Machines A,B,C,D
- Machines run 2400 min/week
- fixed expenses of $5000/week

Modeling Goldratt Problem

Formulation: ***X_p = weekly production of P, X_q = weekly production of Q***

$$
\begin{array}{lll}
\max & (90-45)X_p + (100-40)X_q - 5000 & \textit{Weekly Profit} \\
\text{s.t.} & 15X_p + 10X_q \le 2400 & \textit{Time on Machine A} \\
 & 15X_p + 30X_q \le 2400 & \textit{Time on Machine B} \\
 & 15X_p + 5X_q \le 2400 & \textit{Time on Machine C} \\
 & 15X_p + 5X_q \le 2400 & \textit{Time on Machine D} \\
 & X_p \le 100 & \textit{Max Sales of P} \\
 & X_q \le 50 & \textit{Max Sales of Q}
\end{array}
$$

Solution Approach:

1. Choose (feasible) production quantity of P (X_p) or Q (X_q).
2. Use remaining capacity to make other product.

Unit Profit Approach

Make as much Q as possible because it is highest priced:

$$X_q \leq 50 \qquad (*)$$

$$10X_q \leq 2400 \quad \rightarrow \quad X_q \leq 240 \qquad \textbf{A}$$

$$30X_q \leq 2400 \quad \rightarrow \quad X_q \leq 80 \qquad \textbf{B}$$

$$5X_q \leq 2400 \quad \rightarrow \quad X_q \leq 480 \qquad \textbf{C,D}$$

$$X_p \leq 100$$

$$15X_p + 10(50) \leq 2400 \quad \rightarrow \quad X_p \leq 126.67 \qquad \textbf{A}$$

$$15X_p + 30(50) \leq 2400 \quad \rightarrow \quad X_p \leq 60 \quad (*) \qquad \textbf{B}$$

$$15X_p + 5(50) \leq 2400 \quad \rightarrow \quad X_p \leq 143.33 \qquad \textbf{C,D}$$

$$X_p = 60$$

$$X_q = 50$$

$$Z = 45(60) + 60(50) - 5000 = 700$$

Bottleneck Ratio Approach

Consider bottleneck: If we set $\boldsymbol{X_p}$ =100, $\boldsymbol{X_q}$ =50, we violate capacity constraint

$$15(100) + 10(50) = 2000 \leq 2400 \qquad \textbf{A}$$

$$15(100) + 30(50) = 3000 > 2400 \qquad (*) \quad \textbf{B}$$

$$15(100) + 5(50) = 1750 \leq 2400 \qquad \textbf{C}$$

$$15(100) + 5(50) = 1750 \leq 2400 \qquad \textbf{D}$$

Profit/Unit of Bottleneck Resource ($/minute):

X_p : 45/15 = 3

X_q : 60/30 = 2

so make as much P as possible (i.e., set X_p =100, since this does not violate any of the capacity constraints):

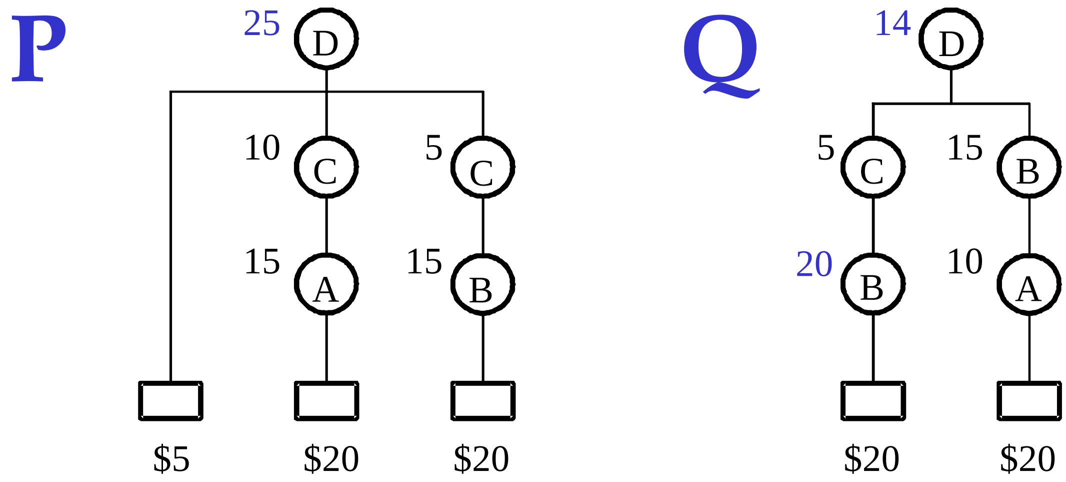
Bottleneck Ratio Approach (cont.)

$$
\begin{array}{llll}
X_q \le 100 & & & \\
15(100) + 10X_q \le 2400 & \rightarrow & X_q \le 90 & \textbf{A} \\
15(100) + 30X_q \le 2400 & \rightarrow & X_q \le 30 \quad (*) & \textbf{B} \\
15(100) + 5X_q \le 2400 & \rightarrow & X_q \le 180 & \textbf{C,D}
\end{array}
$$

$$
\begin{aligned}
X_p &= 100 \\
X_q &= 30 \\
Z &= 45(100) + 60(30) - 5000 = 1300
\end{aligned}
$$

Outcome: This turns out to be the best we can do. But will this approach always work?

Modified Goldratt Problem



Note: *only minor changes to times.*

Product	P	Q
Price	$90	$100
Max Weekly Sales	100	50

- Machines A,B,C,D
- Machines run 2400 min/week
- fixed expenses of $5000/week

Modeling Modified Goldratt Problem

Formulation:

max	$45X_p + 60X_q - 5000$	*Weekly Profit*
s.t.	$15X_p + 10X_q \le 2400$	*Time on Machine A*
	$15X_p + 35X_q \le 2400$	*Time on Machine B*
	$15X_p + 5X_q \le 2400$	*Time on Machine C*
	$25X_p + 15X_q \le 2400$	*Time on Machine D*
	$X_p \le 100$	*Max Sales of P*
	$X_q \le 50$	*Max Sales of Q*

Solution Approach: bottleneck method.

Bottleneck Solution

Find Bottleneck:

$$15(100)+10(50)=2000 \leq 2400 \qquad\qquad \textbf{A}$$

$$15(100)+35(50)=3250 > 2400 \qquad (*) \quad \textbf{B}$$

$$15(100)+5(50)=1750 \leq 2400 \qquad\qquad \textbf{C}$$

$$25(100)+15(50)=3250 > 2400 \qquad (*) \quad \textbf{D}$$

Note: Both B and D are bottlenecks! (Does this seem unrealistic in a world where line balancing is a way of life?)

Possible Solutions

Make as much P as possible:

$$25X_p \le 2400 \quad \rightarrow \quad X_p \le 96$$

$$15(96) + 10X_q \le 2400 \quad \rightarrow \quad X_q \le 96 \qquad \textbf{A}$$

$$15(96) + 35X_q \le 2400 \quad \rightarrow \quad X_q \le 27.43 \qquad \textbf{B}$$

$$15(96) + 5X_q \le 2400 \quad \rightarrow \quad X_q \le 192 \qquad \textbf{C}$$

$$25(96) + 15X_q \le 2400 \quad \rightarrow \quad X_q \le 0 \qquad \textbf{D}$$

$$X_p = 96$$

$$X_q = 0$$

$$Z = 45(96) + 60(0) - 5000 = -680$$

Possible Solutions (cont.)

Make as much Q as possible:

$$35X_q \le 2400 \quad \rightarrow \quad X_q \le 87.5$$

so make $X_q = 50$ (can't sell more than this)

$$15X_p + 10(50) \le 2400 \quad \rightarrow \quad X_p \le 126.67 \quad \textbf{A}$$

$$15X_p + 35(50) \le 2400 \quad \rightarrow \quad X_p \le 43.33 \quad \textbf{B}$$

$$15X_p + 5(50) \le 2400 \quad \rightarrow \quad X_p \le 143.33 \quad \textbf{C}$$

$$25X_p + 15(50) \le 2400 \quad \rightarrow \quad X_p \le 66 \quad \textbf{D}$$

$$X_p = 43.33$$

$$X_q = 50$$

$$Z = 45(43.33) + 60(50) - 5000 = -50$$

Another Solution

Make X_p =73, X_q =37: (Where in the heck did these come from? A model!)

A $15(73)+10(37)=1465\leq 2400$

B $15(73)+35(37)=2390\leq 2400$

C $15(73)+5(37)=1280\leq 2400$

D $25(73)+15(37)=2380\leq 2400$ all constraints satisfied

$$Z=45(73)+60(37)-5000=505$$

Conclusions:

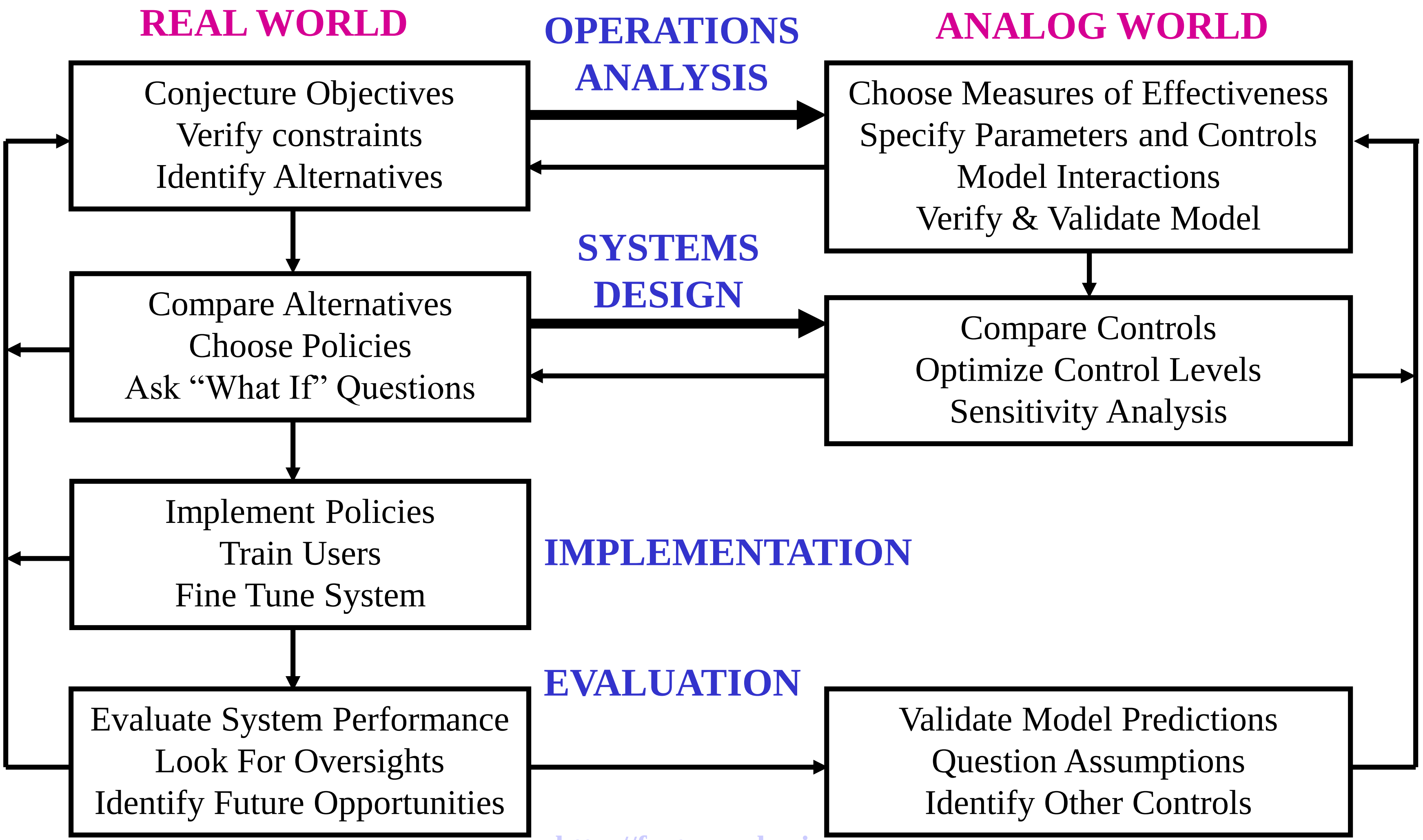
- Modeling matters!
- Beware of simplistic solutions to complex problems!

Systems Analysis

Definition: ***Systems analysis*** is a structured approach to problem-solving that involves

1. Identification of ***objectives*** (what you want to accomplish), *measures* (for comparing alternatives), and ***controls*** (what you can change).
2. Generation of specific ***alternatives***.
3. ***Modeling*** (some form of abstraction from reality to facilitate comparison of alternatives).
4. ***Optimization*** (at least to the extent of ranking alternatives and choosing "best" one).
5. ***Iteration*** (going back through the process as new facets arise).

System Analysis Paradigm



http://factory-physics.com

Hierarchical Objectives